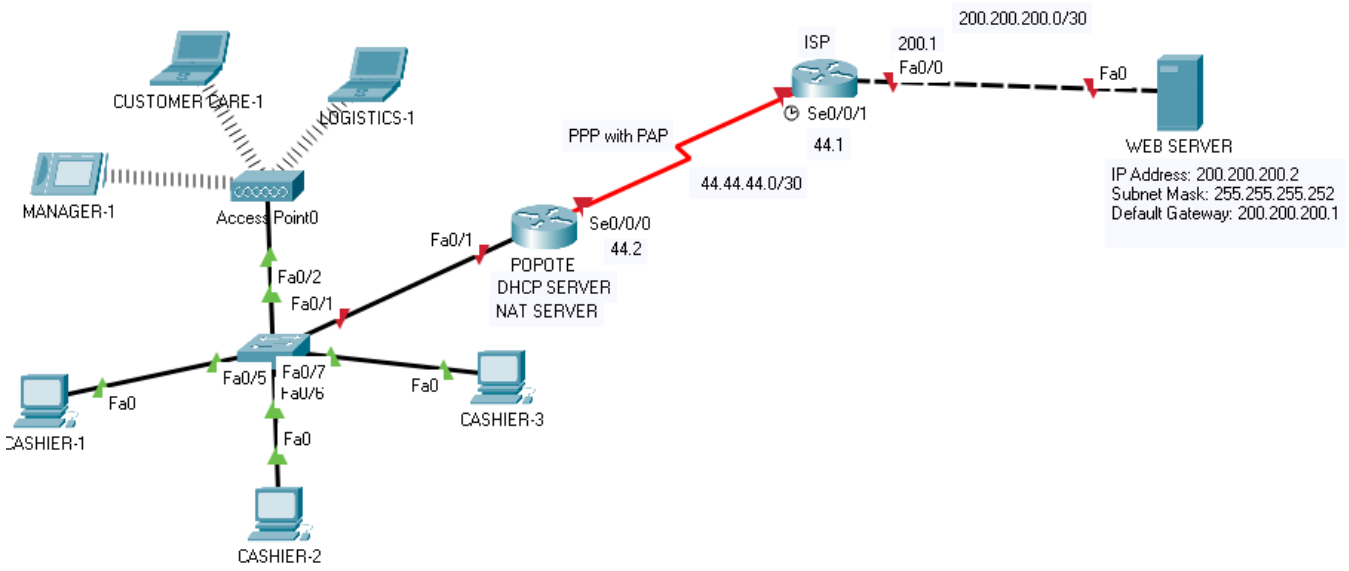
Strathmore UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF INFORMATICS & COMPUTER SCIENCE
ICS 2203: ADVANCED NETWORKING
LAB ASSIGN 4: CONFIGURING A SWITCHED LAN

Instructions

1. This mini-semester project will be done on groups of two
2. You will require about a week and a half to complete the work.
3. You will submit your complete project between **by 10th November 2026, 23:59, EAT**

The Topology



The Scenario

Popote Cosmetics is an upcoming startup with only one branch in the Central Business District (CBD) of one of the East Africam countries. Popote handles cosmetics sales in a physical shop but also operates an ecomerce platform to reach its online clientelle. To support its business operations, Popote Cosmetics has several cashiers on site to handle the walk-in shoppers, three logistic staff to deal with online orders, 2 customer service attandants to handle customer complaints and two managers who also sometimes double up in other roles such as accounting, IT support and supervsion of the employees. The company relies on outsourced delivery services within the city.

The owners of the business anticipate growth and hope to open more outlets in the foreseeable future.

Required:

Use the topology, information in the scenario and the device model table below to perform the tasks that follow

Device Name	Device Model	Location in packet Tracer
Routers	1841	Miscelleneous Folder
Access Point	AP-PT	Wireless Devices
Switch	2960	Switches

Laptops	Wireless Laptop	Miscelleneous Folder
Tablet (Manager-1)	TabletPC-PT	End Devices
PCs	PC-PT	End Devices
Server	Server-PT	End Devices

Device Model Table

Part 1: Preparation

a) Select a private IP address space of your choice to use on the given topology
b) Assign a /24 prefix to whichever IP you have chosen in (a) above
c) From you answer in (a) and prefix in (b) subnet your IP into 4 equal sized subnets
d) Using your answer in (c) above, fill the subnet chart below.

Subnet No.	Subnet Address	Usable Host Addresses (Valid host addresses)	Broadcast Address	Prefix

Table 1

e) Assume that you only need two VLANs at the moment and have grouped the users as follows:
- Administrative staff i.e. the employees in logistics, customer care and managers
- The Cashiers i.e. those handling on site sales

Fill the table below ensuring that you allocate each VLAN a specific subnet

User Group	VLAN Port Association	VLAN ID	VLAN Name	Subnet Address	Prefix
Administrative staff	fa0/2 - fa0/4				
Cashiers	fa0/5 - fa0/24				

Table 2

f) From the topoloigy diagram, all the administrative staff devices will be on the wireless LAN, and will be assigned IP settings dynamically via DHCP, while the IPs for the cashiers PCs will be assigned IP settings by manual configuration.
Determine:
   i. The IP settings that you will configure on the PCs in the Cashiers' VLAN and fill the table below.
   Note: *You will need to refer to Tables 1 and 2 for the specific IPs needed by the cashier's VLAN*

PC	Interface IP Address	Subnet Mask	Default Gateway
Cashier-1			
Cashier-2			
Cashier-3			

Table 3

   ii. The usable host address range for the devices in the administrative Staff VLAN. Fill table below.
   Note: *You will need to refer to Tables 1 and 2*

User Group	Usable Host addresses	Subnet Mask	Default Gateway
Administrative staff			

Table 4

Note*: You will use the data in the tables in (f) for various configurations*

Part 2: Connections

Connect your devices as in the topolgy on page 1. Refer to the device table and the topology on page 1

Part 3: Configure VLANs on Popote VLAN Switch

a) Using your answers on Table 2, configure and name the VLANs on the Popote LAN switch
b) Assign the VLANs to appropriate interfaces. Refer to Table 2
c) Configure Fa0/1 on the switch as trunk
d) ''show intrerface trunk' to verify your VLAN configurations above

Part 4: Configure inter-VLAN routing on the Popote Router

a) Using the data in Tables, 2, 3 and 4 and the topology, identify the router's sub-interface IP addresses and fill the table below

Sub-interface	VLAN_ID	Interface IP Address	Subnet Mask

Table 5

Note: *The router's sub-interface IP address will act as the default gateway to each VLAN on the LAN*

b) Configire the router name
c) Using the data in Table 5, configure inter-VLAN routing on the router
d) Verify the router configurations above and the switch trunk configurations in Part 3

Part 5: Configure DHCP on Popote Router

a) Conduct some research here to get the commands that you would need to enable DHCP service on a Cisco router when the toppology has VLANs.

Note

Your DHCP pool will only include IPs in the administrators' VLAN.
Refer to Tables 2 and 4 for the correct IP range and gateway address information

Part 6: Configure IP addresses on Popote's LAN

a) Configure the Administrative staff PCs to use DHCP to acquire IP addresses
    i. From the wirelss devices' desktop IP settings confirm that the IP, mask and default gateway are correct as per the configuration done on part 5 above
    ii. Ping from the wireless devices to the default gateway. This ping should succeed if PCs' IP settings are correct and router sub-interface configurations are correctly done
b) Configure the Cashier's PCs
    i. Using the data in Table 3, manually configure IP Settings on the Cashiers' PCs
    ii. Ping between the cashiers' PCs to verify connectivity. This ping should succeed if PCs' IP settings are correct and switch VLAN configurations are correctly done
    iii. Ping the default gateway to verify connectivity. This ping should succeed if PCs' IP settings are correct and router sub-interface configurations are correctly done
c) Ping between the wireless devices and the Cashier's PCs. This ping should succeed if all user device IP settings are correct, switch VLAN configurations are correctly done and the inter-VLAN routing is also well configured

Part 7: Configure the WAN Communication

Refer to the topology

a) Configure the IP addresses on the remaining parts of the topology
b) Verify *(using ping)* that the directly connected interfaces can communicate
c) Configure Static routing.
    i. On Popote, configure a default static route to route traffic to unknown destinations to the ISP router *(will conduct some research here)*
    ii. Use the 'show ip route' command to verify that the route appears in the Popote routing table with the code S*
d) Ping between several user devices in the Popote LAN and Web Server. The Pings should succeed.
a) Now check the NAT translation table to verify if you have any private public IP mappings

Task 8: Submit your work for marking

Submission Instructions

- You will submit a word document with your answers for the tables and a packet tracer file.
- Ensure that both files have full information i.e. Admission No. and Names of the team members

Note: ***You may be tasked to present your work if time allows so that you get feedback on the spot***